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CLOUD COMPUTING

C01 - AT3

SHORT ANSWER TEST

1) Define cloud computing and explain four characteristics:

Definition:

Cloud computing is the delivery of computing service such as server storage, database, networking and software through the Internet instead of using local computer.

Four characteristics:

1) On-demand self-service - user can get computing resources whenever required without human interaction

2) Board network access - Cloud services can be accessed through the Internet using mobiles, laptops and other device

3) Resource pooling - cloud provider share computing resources among many user efficiently.

4) Rapid elasticity - Resources can be increased or decreased quickly according to demand.

Cloud computing is flexible, scalable and cost-effective compared with tradional on-primese computing.

) Evolution of cloud Computing:

Cloud computing developed through several stages:

* Hard Evolution.

Computer became smaller, faster and cheaper: this improved computing capacity.

* Mainframe computing

Many user accessed a single powerful mainframe computer through terminals.

* Client - Server computing

Processing was divided between client computers and servers connected through networks.

* Web and software Evolution:

Web application and boowser allowed software to be delivered through the internet.

* Virtualization.

One physical server could run multiple virtual method the improved hardward utilization.

* Modern cloud computing.

Virtualization + high-speed Internet + data centers + automation created modern cloud Platforms.

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③ Role of Server Virtualization:

Server Virtualization divide one physical server into multiple virtual machine (VMs) using a software layer called as hypervisor.

Role in cloud computing:

* Improves server utilization
* Reduce hardware cost
* Allow multiple user to share resources
* Provides flexibility and scalability
* makes resources allocation easier.
* Support quick creation and deletion of virtual machines.

Virtualization vs cloud computing:

Virtualization	cloud computing
Technology	Service delivery model
creates virtual resources	Deliver computing services
can work without Internet	Usually accessed through network
Example: Virtual Machine	Ex - AWS EC2

In simple words:

Virtualization is an enabling technology, while cloud computing uses such technologies to provide services of user.

④ Role of data Centers in cloud Services:

A data center is a facility containing servers, storage system, networking equipment and other infrastructure.

Roles:

* Computing - Provides servers to run application.
* Storage - Store user data and application
* Networking - Connect cloud resources to user through the Internet
* Security - Protects data using firewalls, authentication and monitoring
* Backup - Maintain copies of data to prevent data loss
* Scalability - Provides additional resources when demand.
* Availability - Uses redundant system to keep service running

Data center are the physical foundation of cloud computing they provide the hardware and infrastructure required to deliver cloud services.

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⑤ Iaas, Paas, Saas and Xaas

∴ Cloud computing Provides different services based on user requirements.

Iaas - Infrastructure as a Services:

Provide Virtual Service, Storage and networking through the Internet

Ex- Amazon EC2

② Paas - Platform as a Service:

Provide virtual server storage and networking through the Internet. test deploy applications.

Ex- Google App Engine

③ Saas - Software as a Service:

Provide ready-to-use software through the Internet User need not install or maintain it.

Ex- Google Workspace.

④ Xaas - Anything as a Service:

Provide different IT services through the cloud. Such as database, Storage and security services.

Ex- Amazon RDS

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